

Part A – Identify the ML Problem

Chosen Application: Product Recommendation

Question	Your Answer
1. What is the application?	Product recommendation (e.g., an online shopping app suggesting items to a user)
2. What problem are you trying to solve?	Recommend products that a specific user is likely to want to buy
3. What are the inputs?	User's past purchases, items viewed, items added to cart, ratings given
4. What is the expected output?	A list of recommended products for that user
5. What patterns or relationships might the ML system learn?	Users with similar browsing/purchase habits tend to like similar products; buying one item often leads to buying related items
6. What data would the system need?	Purchase history, browsing history, product categories, ratings/reviews, and similar data from many users

Part B – Traditional Programming vs. Machine Learning

Traditional Programming (Input → Algorithm → Output)

The programmer would need to manually write specific rules, like: "If a user buys a phone, suggest a phone case," or "If a user views 3+ items in the 'shoes' category, show more shoes." Every rule has to be thought of and coded by hand, and the programmer would need to predict every possible situation in advance — which becomes unrealistic as the number of products and users grows.

Machine Learning (Input + Output/Data → ML Algorithm)

Instead of writing rules, the programmer gives the algorithm many examples of past user behavior (inputs) paired with what those users actually bought or liked (outputs). The algorithm studies these examples and learns the pattern on its own — for example, that people who buy running shoes often also buy socks — without anyone explicitly coding that rule.

Part C – Create Your Own Mini Dataset

Dataset: Number of items viewed → Number of items purchased

Input (items viewed)	Output (items purchased)
1	0
2	1
3	1
4	2
5	?

1. **Pattern:** Roughly, purchases increase as views increase — about 1 purchase for every 2 items viewed.
2. **Missing output:** Following the pattern, 5 items viewed → likely **2 purchases**.
3. **Mathematical relationship:** $\text{Output} \approx \text{Input} \div 2$ (rounded down), or $\text{output} = \text{floor}(\text{input} / 2)$.
4. **How an ML algorithm could learn this:** By looking at the paired examples (1→0, 2→1, 3→1, 4→2), the algorithm would test different possible relationships and adjust until it finds one that best fits all the examples — essentially discovering the "divide by 2" pattern on its own, without being told the formula.

Part D – Explain the Training Process

A Machine Learning algorithm learns by being shown many examples of inputs paired with their correct outputs. During training, it repeatedly compares its own guesses to the actual correct outputs and adjusts itself to reduce the difference between them. Over many rounds of this, it gradually figures out the underlying relationship connecting inputs to outputs. Once trained, it can apply that learned relationship to new inputs it hasn't seen before, producing a prediction. In short: Input → Output examples are used for Training → the algorithm learns the Relationship → and then it can make a Prediction on new data.

Reflection

Why is learning from data useful instead of manually programming every possible rule?

Manually programming every rule is only realistic for simple, predictable tasks. For complex problems — like recommending products among millions of items, or translating language — there are far too many possible situations and exceptions for a person to think of and code by

hand. Learning from data lets the system figure out patterns directly from real examples, including patterns a human might not even notice, and it can keep improving as more data becomes available. This makes ML far more scalable and adaptable than trying to hand-write rules for every case.